

Entry: 1

Item: 2

Question:

Let's examine a simple program, "loop.s". First, just read and understand it. Then, run it with these arguments (./x86.py -p

Answer:

We need to run like this: ./x86.py -p loop.s -t 1 -i 100 -R dx -c Here is how it's executed. It simply subtracts 1 from %dx re

Sample answer:

```
dx Thread 0
0
-1 1000 sub $1,%dx
-1 1001 test $0,%dx
-1 1002 jgte .top
-1 1003 halt
```

Sample criteria:

The program code or text explanation can explain the process of value changing from 0 to -1. Correct (16 points);
If only answer the final value is -1 . Partically correct (4 points); If only asnser the "change range is from 0 to -1 (8 p

Full points:

16

Score 1:

16

Score 2:

16

Score 3:

16

Score outlier:

6

Entry: 2

Item: 2

Question:

Let's examine a simple program, "loop.s". First, just read and understand it. Then, run it with these arguments (./x86.py -p

Answer:

%dx will be -1

Sample answer:

```
dx Thread 0
0
-1 1000 sub $1,%dx
-1 1001 test $0,%dx
-1 1002 jgte .top
-1 1003 halt
```

Sample criteria:

The program code or text explanation can explain the process of value changing from 0 to -1. Correct (16 points);
If only answer the final value is -1 . Partically correct (4 points); If only asnser the "change range is from 0 to -1 (8 p

Full points:

16

Score 1:

8.0

Score 2:

4.0

Score 3:

4

Entry: 3

Item: 2

Question:

Let's examine a simple program, "loop.s". First, just read and understand it. Then, run it with these arguments (./x86.py -p

Answer:

%dx will be change from 0 to -1

Sample answer:

```
dx Thread 0
0
-1 1000 sub $1,%dx
-1 1001 test $0,%dx
-1 1002 jgte .top
-1 1003 halt
```

Sample criteria:

The program code or text explanation can explain the process of value changing from 0 to -1. Correct (16 points);
If only answer the final value is -1 . Partically correct (4 points); If only asnsr the "change range is from 0 to -1 (8 p

Full points:

16

Score 1:

8

Score 2:

8

Score 3:

8

Entry: 4

Item: 2

Question:

Let's examine a simple program, `loop.s`. First, just read and understand it. Then, run it with these arguments:

Answer:

`%dx` will be 0 before the run, and -1 during the run.

Sample answer:

```
dx Thread 0
0
-1 1000 sub $1,%dx
-1 1001 test $0,%dx
-1 1002 jgte .top
-1 1003 halt
```

Sample criteria:

The program code or text explanation can explain the process of value changing from 0 to -1. Correct (16 points);
If only answer the final value is -1 . Partically correct (4 points); If only answer the "change range is from 0 to -1 (8 points)

Full points:

16

Score 1:

8.0

Score 2:

8.0

Score 3:

8

Entry: 5

Item: 2

Question:

Let's examine a simple program, \

Answer:

0, -1, -1, -1, -1

Sample answer:

```
dx Thread 0
0
-1 1000 sub $1,%dx
-1 1001 test $0,%dx
-1 1002 jgte .top
-1 1003 halt
```

Sample criteria:

The program code or text explanation can explain the process of value changing from 0 to -1. Correct (16 points);
If only answer the final value is -1 . Partically correct (4 points); If only asnser the "change range is from 0 to -1 (8 p

Full points:

16

Score 1:

12

Score 2:

12

Score 3:

12

Score outlier:

4

Entry: 6

Item: 2

Question:

Let's examine a simple program, "loop.s". First, just read and understand it. Then, run it with these arguments (./x86.py -p

Answer:

dx should be -1.

Sample answer:

```
dx Thread 0
0
-1 1000 sub $1,%dx
-1 1001 test $0,%dx
-1 1002 jgte .top
-1 1003 halt
```

Sample criteria:

The program code or text explanation can explain the process of value changing from 0 to -1. Correct (16 points);
If only answer the final value is -1 . Partically correct (4 points); If only asnser the "change range is from 0 to -1 (8 p

Full points:

16

Score 1:

4.0

Score 2:

4

Score 3:

4

Entry: 7

Item: 2

Question:

Let's examine a simple program, "loop.s". First, just read and understand it. Then, run it with these arguments (./x86.py -p

Answer:

%dx will be -1. dx Thread 0 0 1 1000 sub \$1,%dx 1 1001 test \$0,%dx 1 1002 jgte .top 1 1003 halt

Sample answer:

```
dx Thread 0
0
-1 1000 sub $1,%dx
-1 1001 test $0,%dx
-1 1002 jgte .top
-1 1003 halt
```

Sample criteria:

The program code or text explanation can explain the process of value changing from 0 to -1. Correct (16 points);
If only answer the final value is -1 . Partically correct (4 points); If only asnser the "change range is from 0 to -1 (8 p

Full points:

16

Score 1:

16

Score 2:

16

Score 3:

16

Entry: 8

Item: 2

Question:

Let's examine a simple program, "loop.s". First, just read and understand it. Then, run it with these arguments (./x86.py -p

Answer:

the %dx register will see values from 0 to -1 during the run.

Sample answer:

```
dx Thread 0
0
-1 1000 sub $1,%dx
-1 1001 test $0,%dx
-1 1002 jgte .top
-1 1003 halt
```

Sample criteria:

The program code or text explanation can explain the process of value changing from 0 to -1. Correct (16 points);
If only answer the final value is -1 . Partically correct (4 points); If only asnser the "change range is from 0 to -1 (8 p

Full points:

16

Score 1:

8

Score 2:

8

Score 3:

8

Entry: 9

Item: 2

Question:

Let's examine a simple program, "loop.s". First, just read and understand it. Then, run it with these arguments (./x86.py -p loop.s -t 1 -i 100 -R dx). This spe

Answer:

%dx is 0 at first Then it sub 1 %dx is -1 Then it is still -1

Sample answer:

```
dx Thread 0
0
-1 1000 sub $1,%dx
-1 1001 test $0,%dx
-1 1002 jgte .top
-1 1003 halt
```

Sample criteria:

The program code or text explanation can explain the process of value changing from 0 to -1. Correct (16 points);
If only answer the final value is -1 . Partically correct (4 points); If only asnser the "change range is from 0 to -1 (8 p

Full points:

16

Score 1:

16.0

Score 2:

16.0

Score 3:

16.0

Score outlier:

5

Entry: 10

Item: 2

Question:

Let's examine a simple program, "loop.s". First, just read and understand it. Then, run it with these arguments:

Answer:

-1

Sample answer:

```
dx Thread 0
0
-1 1000 sub $1,%dx
-1 1001 test $0,%dx
-1 1002 jgte .top
-1 1003 halt
```

Sample criteria:

The program code or text explanation can explain the process of value changing from 0 to -1. Correct (16 points);
If only answer the final value is -1 . Partically correct (4 points); If only asnser the "change range is from 0 to -1 (8 points)

Full points:

16

Score 1:

16.0

Score 2:

16.0

Score 3:

16.0

Entry: 11

Item: 2

Question:

Let's examine a simple program, \

Answer:

%dx will be -1

Sample answer:

```
dx Thread 0
0
-1 1000 sub $1,%dx
-1 1001 test $0,%dx
-1 1002 jgte .top
-1 1003 halt
```

Sample criteria:

The program code or text explanation can explain the process of value changing from 0 to -1. Correct (16 points);
If only answer the final value is -1 . Partically correct (4 points); If only asnsr the "change range is from 0 to -1 (8 p

Full points:

16

Score 1:

8.0

Score 2:

4.0

Score 3:

4

Score outlier:

16

Entry: 12

Item: 2

Question:

Let's examine a simple program, "loop.s". First, just read and understand it. Then, run it with these arguments (./x86.py -p

Answer:

0, -1, -1, -1, -1

Sample answer:

```
dx Thread 0
0
-1 1000 sub $1,%dx
-1 1001 test $0,%dx
-1 1002 jgte .top
-1 1003 halt
```

Sample criteria:

The program code or text explanation can explain the process of value changing from 0 to -1. Correct (16 points);
If only answer the final value is -1 . Partically correct (4 points); If only asnser the "change range is from 0 to -1 (8 p

Full points:

16

Score 1:

16

Score 2:

16

Score 3:

12

Entry: 13

Item: 2

Question:

Let's examine a simple program, "loop.s". First, just read and understand it. Then, run it with these arguments (./x86.py -p

Answer:

Initially dx will be 0 and it will become -1 after running the code

Sample answer:

```
dx Thread 0
0
-1 1000 sub $1,%dx
-1 1001 test $0,%dx
-1 1002 jgte .top
-1 1003 halt
```

Sample criteria:

The program code or text explanation can explain the process of value changing from 0 to -1. Correct (16 points);
If only answer the final value is -1 . Partically correct (4 points); If only asnser the "change range is from 0 to -1 (8 p

Full points:

16

Score 1:

16

Score 2:

16

Score 3:

8

Entry: 14

Item: 2

Question:

Let's examine a simple program, "loop.s". First, just read and understand it. Then, run it with these arguments (./x86.py -p

Answer:

-1

Sample answer:

```
dx Thread 0
0
-1 1000 sub $1,%dx
-1 1001 test $0,%dx
-1 1002 jgte .top
-1 1003 halt
```

Sample criteria:

The program code or text explanation can explain the process of value changing from 0 to -1. Correct (16 points);
If only answer the final value is -1 . Partically correct (4 points); If only asnser the "change range is from 0 to -1 (8 p

Full points:

16

Score 1:

4

Score 2:

4

Score 3:

4

Entry: 15

Item: 2

Question:

Let's examine a simple program, "loop.s". First, just read and understand it. Then, run it with these arguments (./x86.py -p

Answer:

-1

Sample answer:

```
dx Thread 0
0
-1 1000 sub $1,%dx
-1 1001 test $0,%dx
-1 1002 jgte .top
-1 1003 halt
```

Sample criteria:

The program code or text explanation can explain the process of value changing from 0 to -1. Correct (16 points);
If only answer the final value is -1 . Partically correct (4 points); If only asnser the "change range is from 0 to -1 (8 p

Full points:

16

Score 1:

4

Score 2:

4

Score 3:

4

Entry: 16

Item: 2

Question:

Let's examine a simple program, "loop.s". First, just read and understand it. Then, run it with these arguments (./x86.py -p

Answer:

%dx is -1 during the run

Sample answer:

```
dx Thread 0
0
-1 1000 sub $1,%dx
-1 1001 test $0,%dx
-1 1002 jgte .top
-1 1003 halt
```

Sample criteria:

The program code or text explanation can explain the process of value changing from 0 to -1. Correct (16 points);
If only answer the final value is -1 . Partically correct (4 points); If only asnser the "change range is from 0 to -1 (8 p

Full points:

16

Score 1:

4

Score 2:

4

Score 3:

4

Entry: 17

Item: 2

Question:

Let's examine a simple program, "loop.s". First, just read and understand it. Then, run it with these arguments

Answer:

dx = -1.

Sample answer:

```
dx Thread 0
0
-1 1000 sub $1,%dx
-1 1001 test $0,%dx
-1 1002 jgte .top
-1 1003 halt
```

Sample criteria:

The program code or text explanation can explain the process of value changing from 0 to -1. Correct (16 points);
If only answer the final value is -1 . Partically correct (4 points); If only asnser the "change range is from 0 to -1 (8 p

Full points:

16

Score 1:

4.0

Score 2:

4.0

Score 3:

4

Entry: 18

Item: 2

Question:

Let's examine a simple program, "loop.s". First, just read and understand it. Then, run it with these arguments (./x86.py -p

Answer:

The value of %dx will be -1.

Sample answer:

```
dx Thread 0
0
-1 1000 sub $1,%dx
-1 1001 test $0,%dx
-1 1002 jgte .top
-1 1003 halt
```

Sample criteria:

The program code or text explanation can explain the process of value changing from 0 to -1. Correct (16 points);
If only answer the final value is -1 . Partically correct (4 points); If only asnser the "change range is from 0 to -1 (8 p

Full points:

16

Score 1:

4

Score 2:

4

Score 3:

4

Entry: 19

Item: 2

Question:

Let's examine a simple program, "loop.s". First, just read and understand it. Then, run it with these arguments (./x86.py -p

Answer:

0 in %dx in the beginning. In sub \$1, %dx, %dx become -1. In test \$0, %dx, -1 in %dx is smaller than 0. %dx remains und

Sample answer:

```
dx Thread 0
0
-1 1000 sub $1,%dx
-1 1001 test $0,%dx
-1 1002 jgte .top
-1 1003 halt
```

Sample criteria:

The program code or text explanation can explain the process of value changing from 0 to -1. Correct (16 points);
If only answer the final value is -1 . Partically correct (4 points); If only asnser the "change range is from 0 to -1 (8 p

Full points:

16

Score 1:

16

Score 2:

16

Score 3:

16

Entry: 20

Item: 2

Question:

Let's examine a simple program, "loop.s". First, just read and understand it. Then, run it with these arguments (./x86.py -p

Answer:

%dx will become 0 and -1 during the run.

Sample answer:

```
dx Thread 0
0
-1 1000 sub $1,%dx
-1 1001 test $0,%dx
-1 1002 jgte .top
-1 1003 halt
```

Sample criteria:

The program code or text explanation can explain the process of value changing from 0 to -1. Correct (16 points);
If only answer the final value is -1 . Partically correct (4 points); If only asnser the "change range is from 0 to -1 (8 p

Full points:

16

Score 1:

12

Score 2:

8

Score 3:

8

Entry: 21

Item: 2

Question:

Let's examine a simple program, \u201cloop.s\u201d. First, just read and understand it. Then, run it with these arguments:

Answer:

dx Thread 0 0 -1 1000 sub \$1,%dx -1 1001 test \$0,%dx -1 1002 jgte .top -1 1003 halt

Sample answer:

```
dx Thread 0
0
-1 1000 sub $1,%dx
-1 1001 test $0,%dx
-1 1002 jgte .top
-1 1003 halt
```

Sample criteria:

The program code or text explanation can explain the process of value changing from 0 to -1. Correct (16 points);
If only answer the final value is -1 . Partically correct (4 points); If only asnser the "change range is from 0 to -1 (8 points)

Full points:

16

Score 1:

16

Score 2:

16

Score 3:

16

Entry: 22

Item: 2

Question:

Let's examine a simple program, "loop.s". First, just read and understand it. Then, run it with these arguments (./x86.py -p

Answer:

0 1 1 1 1

Sample answer:

```
dx Thread 0
0
-1 1000 sub $1,%dx
-1 1001 test $0,%dx
-1 1002 jgte .top
-1 1003 halt
```

Sample criteria:

The program code or text explanation can explain the process of value changing from 0 to -1. Correct (16 points);
If only answer the final value is -1 . Partically correct (4 points); If only asnser the "change range is from 0 to -1 (8 p

Full points:

16

Score 1:

0

Score 2:

0

Score 3:

0

Entry: 23

Item: 2

Question:

Let's examine a simple program, \

Answer:

python2 x86.py -p loop.s -t 1 -i 100 -R dx -c ARG seed 0 ARG numthreads 1 ARG program loop.s ARG interrupt frequen

Sample answer:

```
dx Thread 0
0
-1 1000 sub $1,%dx
-1 1001 test $0,%dx
-1 1002 jgte .top
-1 1003 halt
```

Sample criteria:

The program code or text explanation can explain the process of value changing from 0 to -1. Correct (16 points);
If only answer the final value is -1 . Partically correct (4 points); If only asnser the "change range is from 0 to -1 (8 p

Full points:

16

Score 1:

16

Score 2:

16

Score 3:

16

Entry: 24

Item: 2

Question:

Let's examine a simple program, "loop.s". First, just read and understand it. Then, run it with these arguments (./x86.py -p

Answer:

-1

Sample answer:

```
dx Thread 0
0
-1 1000 sub $1,%dx
-1 1001 test $0,%dx
-1 1002 jgte .top
-1 1003 halt
```

Sample criteria:

The program code or text explanation can explain the process of value changing from 0 to -1. Correct (16 points);
If only answer the final value is -1 . Partically correct (4 points); If only asnser the "change range is from 0 to -1 (8 p

Full points:

16

Score 1:

4.0

Score 2:

4

Score 3:

4

Entry: 25

Item: 2

Question:

Let's examine a simple program, \

Answer:

```
python2 x86.py -p loop.s -t 1 -i 100 -R dx -c dx
```

Thread 0

0

```
-1 1000 sub $1,%dx
```

```
-1 1001 test $0,%dx
```

```
-1 1002 jgte .top
```

```
-1 1003 halt
```

The %dx decrease by 1 during the run.

Sample answer:

dx Thread 0

0

```
-1 1000 sub $1,%dx
```

```
-1 1001 test $0,%dx
```

```
-1 1002 jgte .top
```

```
-1 1003 halt
```

Sample criteria:

The program code or text explanation can explain the process of value changing from 0 to -1. Correct (16 points);

If only answer the final value is -1 . Partically correct (4 points); If only asnser the "change range is from 0 to -1 (8 p

Full points:

16

Score 1:

16

Score 2:

16

Score 3:

16

Entry: 26

Item: 2

Question:

Let's examine a simple program, `loop.s`. First, just read and understand it. Then, run it with these arguments:

Answer:

`dx Thread 0 0 -1 1000 sub $1,%dx -1 1001 test $0,%dx -1 1002 jgte .top -1 1003 halt` Sub instruction substitutes 1 from the

Sample answer:

```
dx Thread 0
0
-1 1000 sub $1,%dx
-1 1001 test $0,%dx
-1 1002 jgte .top
-1 1003 halt
```

Sample criteria:

The program code or text explanation can explain the process of value changing from 0 to -1. Correct (16 points);
If only answer the final value is -1 . Partically correct (4 points); If only answer the "change range is from 0 to -1 (8 points)

Full points:

16

Score 1:

16.0

Score 2:

16.0

Score 3:

16

Entry: 27

Item: 2

Question:

Let's examine a simple program, "loop.s". First, just read and understand it. Then, run it with these arguments:

Answer:

dx = 0 initially
dx = -1 after sub \$1,%dx

Sample answer:

```
dx Thread 0
0
-1 1000 sub $1,%dx
-1 1001 test $0,%dx
-1 1002 jgte .top
-1 1003 halt
```

Sample criteria:

The program code or text explanation can explain the process of value changing from 0 to -1. Correct (16 points);
If only answer the final value is -1 . Partically correct (4 points); If only asnser the "change range is from 0 to -1 (8 points)

Full points:

16

Score 1:

16.0

Score 2:

16.0

Score 3:

16

Entry: 28

Item: 2

Question:

Let's examine a simple program, "loop.s". First, just read and understand it. Then, run it with these arguments (./x86.py -p

Answer:

The value of %dx is -1 after the instruction has run.

Sample answer:

```
dx Thread 0
0
-1 1000 sub $1,%dx
-1 1001 test $0,%dx
-1 1002 jgte .top
-1 1003 halt
```

Sample criteria:

The program code or text explanation can explain the process of value changing from 0 to -1. Correct (16 points);
If only answer the final value is -1 . Partically correct (4 points); If only asnsr the "change range is from 0 to -1 (8 p

Full points:

16

Score 1:

4

Score 2:

4

Score 3:

4

Entry: 29

Item: 2

Question:

Let's examine a simple program, "loop.s". First, just read and understand it. Then, run it with these arguments (./x86.py -p

Answer:

dx will be -1 during the run. Dx starts as 0, then the sub instruction subtracts 1 from dx, making it -1. Then in the next ins

Sample answer:

```
dx Thread 0
0
-1 1000 sub $1,%dx
-1 1001 test $0,%dx
-1 1002 jgte .top
-1 1003 halt
```

Sample criteria:

The program code or text explanation can explain the process of value changing from 0 to -1. Correct (16 points);
If only answer the final value is -1 . Partically correct (4 points); If only asnser the "change range is from 0 to -1 (8 p

Full points:

16

Score 1:

16

Score 2:

16

Score 3:

16

Entry: 30

Item: 2

Question:

Let's examine a simple program,

"loop.s". First, just read and understand it. Then, run it with these arguments (./x86.py -p loop.s -t 1 -i 100 -R dx). This spe

Answer:

At first the dx is 0, then it becomes a few -1.

Sample answer:

dx Thread 0

0

-1 1000 sub \$1,%dx

-1 1001 test \$0,%dx

-1 1002 jgte .top

-1 1003 halt

Sample criteria:

The program code or text explanation can explain the process of value changing from 0 to -1. Correct (16 points);

If only answer the final value is -1. Partially correct (4 points); If only answer the "change range is from 0 to -1 (8 p

Full points:

16

Score 1:

8.0

Score 2:

8

Score 3:

8

Entry: 31

Item: 2

Question:

Let's examine a simple program,

"loop.s". First, just read and understand it. Then, run it with these arguments (./x86.py -p loop.s -t 1 -i 100 -R dx). This spe

Answer:

dx Thread 0

0

-1 1000 sub \$1,%dx

-1 1001 test \$0,%dx

-1 1002 jgte .top

-1 1003 halt

Sample answer:

dx Thread 0

0

-1 1000 sub \$1,%dx

-1 1001 test \$0,%dx

-1 1002 jgte .top

-1 1003 halt

Sample criteria:

The program code or text explanation can explain the process of value changing from 0 to -1. Correct (16 points);

If only answer the final value is -1 . Partically correct (4 points); If only asnser the "change range is from 0 to -1 (8 p

Full points:

16

Score 1:

12

Score 2:

16

Score 3:

16

Score outlier:

2

Entry: 32

Item: 2

Question:

Let's examine a simple program, "loop.s". First, just read and understand it. Then, run it with these arguments (./x86.py -p

Answer:

%dx turns -1 from 0 on the sub instruction. The test instruction sees that %dx is smaller than 0 and doesn't execute the ju

Sample answer:

```
dx Thread 0
0
-1 1000 sub $1,%dx
-1 1001 test $0,%dx
-1 1002 jgte .top
-1 1003 halt
```

Sample criteria:

The program code or text explanation can explain the process of value changing from 0 to -1. Correct (16 points);
If only answer the final value is -1 . Partically correct (4 points); If only asnser the "change range is from 0 to -1 (8 p

Full points:

16

Score 1:

16.0

Score 2:

16.0

Score 3:

16

Entry: 33

Item: 2

Question:

Let's examine a simple program, "loop.s". First, just read and understand it. Then, run it with these arguments:

Answer:

dx: 0 0 0 0 0

Sample answer:

dx Thread 0

0

-1 1000 sub \$1,%dx

-1 1001 test \$0,%dx

-1 1002 jgte .top

-1 1003 halt

Sample criteria:

The program code or text explanation can explain the process of value changing from 0 to -1. Correct (16 points);

If only answer the final value is -1 . Partically correct (4 points); If only asnser the "change range is from 0 to -1 (8 points)

Full points:

16

Score 1:

0

Score 2:

0

Score 3:

0

Entry: 34

Item: 2

Question:

Same code, different flags: (./x86.py -p loop.s -t 2 -i 100 -a dx=3,dx=3 -R dx). This specifies two threads, and initializes ea

Answer:

The first thread starts with a dx value of 3, which decreases by 1 after each iteration until it reaches -1. Then, the second

Sample answer:

```
dx Thread 0
0
-1 1000 sub $1,%dx
-1 1001 test $0,%dx
-1 1002 jgte .top
-1 1003 halt
```

Sample criteria:

The program code or text explanation can explain the process of value changing from 0 to -1. Correct (16 points);
If only answer the final value is -1 . Partically correct (4 points); If only asnser the "change range is from 0 to -1 (8 p

Full points:

16

Score 1:

16.0

Score 2:

16.0

Score 3:

16.0

Entry: 35

Item: 2

Question:

Let's examine a simple program, `loop.s`. First, just read and understand it. Then, run it with these arguments:

Answer:

`dx 0 -1 -1 -1 -1`

Sample answer:

```
dx Thread 0
0
-1 1000 sub $1,%dx
-1 1001 test $0,%dx
-1 1002 jgte .top
-1 1003 halt
```

Sample criteria:

The program code or text explanation can explain the process of value changing from 0 to -1. Correct (16 points);
If only answer the final value is -1. Partially correct (4 points); If only answer the "change range is from 0 to -1 (8 points)

Full points:

16

Score 1:

16

Score 2:

16

Score 3:

12

Entry: 36

Item: 2

Question:

Let

{^'}s examine a simple program,

{^'}loop.s■extbf{^'}. First, just read and understand it. Then, run it with these arguments (./x86.py -p loop.s -t

Answer:

dx Thread 0 0 1 1000 sub \$1, %dx 1 1001 test \$0, %dx 1 1002 jgte .top 1 1003 halt

Sample answer:

dx Thread 0

0

-1 1000 sub \$1,%dx

-1 1001 test \$0,%dx

-1 1002 jgte .top

-1 1003 halt

Sample criteria:

The program code or text explanation can explain the process of value changing from 0 to -1. Correct (16 points);

If only answer the final value is -1 . Partically correct (4 points); If only asnsrer the "change range is from 0 to -1 (8 p

Full points:

16

Score 1:

0

Score 2:

0

Score 3:

0

Entry: 37

Item: 2

Question:

Let's examine a simple program,

"loop.s". First, just read and understand it. Then, run it with these arguments (./x86.py -p loop.s -t 1 -i 100 -R dx). This sp

Answer:

%dx will be change from 0 to -1

Sample answer:

dx Thread 0

0

-1 1000 sub \$1,%dx

-1 1001 test \$0,%dx

-1 1002 jgte .top

-1 1003 halt

Sample criteria:

The program code or text explanation can explain the process of value changing from 0 to -1. Correct (16 points);

If only answer the final value is -1 . Partically correct (4 points); If only asnsr the "change range is from 0 to -1 (8 p

Full points:

16

Score 1:

8.0

Score 2:

8.0

Score 3:

8

Entry: 38

Item: 2

Question:

Let's examine a simple program,

"loop.s". First, just read and understand it. Then, run it with these arguments (./x86.py -p loop.s -t 1 -i 100 -R dx). This sp

Answer:

The value is -1.

Sample answer:

dx Thread 0

0

-1 1000 sub \$1,%dx

-1 1001 test \$0,%dx

-1 1002 jgte .top

-1 1003 halt

Sample criteria:

The program code or text explanation can explain the process of value changing from 0 to -1. Correct (16 points);

If only answer the final value is -1 . Partically correct (4 points); If only asnser the "change range is from 0 to -1 (8 p

Full points:

16

Score 1:

4

Score 2:

4

Score 3:

4

Entry: 39

Item: 2

Question:

Let's examine a simple program,

"loop.s". First, just read and understand it. Then, run it with these arguments (./x86.py -p loop.s -t 1 -i 100 -R dx). This sp

Answer:

-1

Sample answer:

dx Thread 0

0

-1 1000 sub \$1,%dx

-1 1001 test \$0,%dx

-1 1002 jgte .top

-1 1003 halt

Sample criteria:

The program code or text explanation can explain the process of value changing from 0 to -1. Correct (16 points);

If only answer the final value is -1 . Partically correct (4 points); If only asnser the "change range is from 0 to -1 (8 p

Full points:

16

Score 1:

4

Score 2:

4

Score 3:

4

Entry: 40

Item: 2

Question:

Let's examine a simple program, `loop.s`. First, just read and understand it. Then, run it with these arguments:

Answer:

%dx will be

-1. Since the register is 0 in the beginning and subtract by 1. The value is -1 which is smaller than 0 so the program is terminated.

Sample answer:

dx Thread 0

0

-1 1000 sub \$1,%dx

-1 1001 test \$0,%dx

-1 1002 jgte .top

-1 1003 halt

Sample criteria:

The program code or text explanation can explain the process of value changing from 0 to -1. Correct (16 points);

If only answer the final value is -1 . Partically correct (4 points); If only answer the "change range is from 0 to -1 (8 points)

Full points:

16

Score 1:

16

Score 2:

16

Score 3:

16.0